

Dataanalyse

API – recap og Eurostat

1. Opvarmning – myLogReg
2. Strings - stringr
3. eurostat / restapi fra GitHub
 1. De unge mødre – Video
 2. De unge mødre - Tyskland



THE FUNCTIONS

1. Downloading and filtering the list of available datasets
 1. **search_eurostat_toc()** This function makes it possible to search through the different tables for a word or a phrase.
 2. **get_eurostat_toc()** This function downloads all the possible tables available.
2. Downloading and searching in the metadata
 1. **get_eurostat_dsd()** returns the Data Structure Definition (DSD) of a given dataset containing the possible dimensions and values with their labels.
 2. **search_eurostat_dsd()** function provides the facility to search for phrase, pattern and regular expressions in the DSD and returns the rows of the DSD where the search string(s) found
3. Retrieving full or partial datasets.
 1. **get_eurostat_data()** function retrieves a data table which can be labeled using the labels from the DSD. The table can contain the whole datasets or only part of it if filters are applied
 2. **get_eurostat_raw()** function downloads the full data table as it is either using the TSV format (default) or the SDMX format keeping all the column names and rows as it is in the original files



Eksempel fra <https://github.com/eurostat/restatapi>

```
dt<-get_eurostat_data("bop_its6_det",  
  filters=list(bop_item="SC",  
               partner="EXT_EU27_2020",  
               geo=c("EU27_2020","HU"),  
               stk_flow="BAL"),  
  date_filter=2020,  
  label=TRUE,  
  name=FALSE)
```



Access Eurostat data with eurostat::cheat sheet

Search and download

Data in the Eurostat database is stored in tables. Each table has an identifier, a short table_code, and a description (e.g. tps00199 - Total fertility rate).

Key eurostat functions allow to find the table_code, download the eurostat table and polish labels in the table.

Find the table code

The `search_eurostat(pattern,...)` function scans the directory of Eurostat tables and returns codes and descriptions of tables that match pattern.

```
library("eurostat")
query <- search_eurostat(pattern = "fertility rate",
                          type = "table", fixed = FALSE)

query[,1:2]
## title                                code
## <chr>                                <chr>
## Total fertility rate by NUTS 2 region tgs00100
## Total fertility rate                  tps00199
## Total fertility rate by NUTS 2 region tgs00100
```

Download the table

The `get_eurostat(id, time_format = "date", filters = "none", type = "code", cache = TRUE,...)` function downloads the requested table from the Eurostat bulk download facility or from The Eurostat Web Services JSON API (if filters are defined). Downloaded data is cached (if cache=TRUE). Additional arguments define how to read the time column (time_format) and if table dimensions shall be kept as codes or converted to labels (type).

```
ct <- c("AT", "BE", "BG", "CH", "CY", "CZ", "DE", "DK", "EE", "EL", "ES",
        "FI", "FR", "HR", "HU", "IE", "IS", "IT", "LI", "LT", "LU", "LV",
        "MT", "NL", "NO", "PL", "PT", "RO", "SE", "SI", "SK", "UK")
dat <- get_eurostat(id="tps00199", time_format="num",
                   filters = list(geo = ct))

dat[1:2,]
## indic_de geo    time values
## TOTFERT AT    2006    1.41
## TOTFERT AT    2007    1.38
```

Add labels

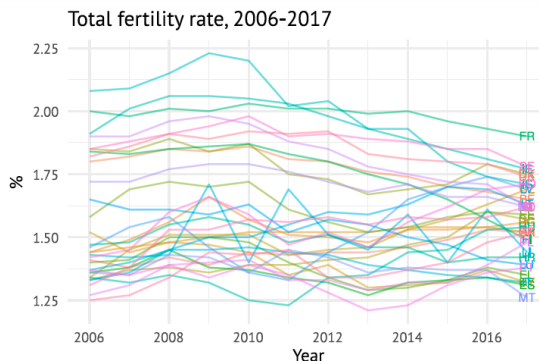
The `label_eurostat(x, lang = "en",...)` gets definitions for Eurostat codes and replace them with labels in given language ('en', 'fr' or 'de')

```
dat <- label_eurostat(dat)
dat[1:3,]
## indic_de geo    time values
## <dbl>    <dbl> <dbl> <dbl>
## Total fertility rate Andorra 2006 1.24
## Total fertility rate Albania 2006 1.67
## Total fertility rate Armenia 2006 1.34
```

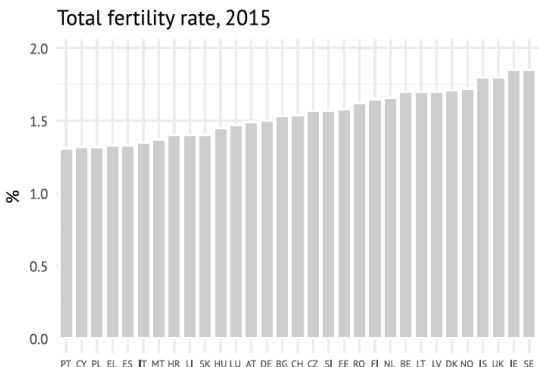
eurostat and plots

The `get_eurostat()` function returns tibbles in the long format. Packages dplyr and tidyr are well suited to transform these objects. The `ggplot2` -package is well suited to plot these objects.

```
dat <- get_eurostat(id="tps00199", filters = list(geo = ct))
library(ggplot2)
library(dplyr)
ggplot(dat,
       aes(x = time, y = values, color = geo, label = geo)) +
  geom_line(alpha = .5) +
  geom_text(data = dat %>% group_by(geo) %>%
            filter(time == max(time)),
            size = 2.6) +
  theme(legend.position = "none") +
  labs(title = "Total fertility rate, 2006-2017",
       x = "Year", y = "%")
```



```
dat_2015 <- dat %>%
  filter(time == "2015-01-01")
ggplot(dat_2015, aes(x = reorder(geo, values), y = values)) +
  geom_col(color = "white", fill = "grey80") +
  theme(axis.text.x = element_text(size = 6)) +
  labs(title = "Total fertility rate, 2015",
       y = "%", x = NULL)
```



eurostat and maps

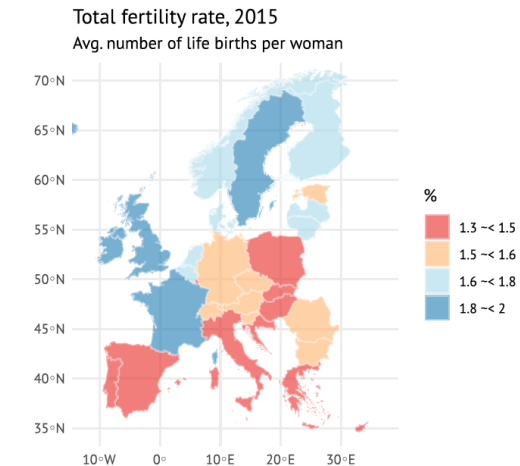
There are two functions to work with geospatial data from GISCO. The `get_eurostat_geospatial()` returns spatial data as sf-object. Object can be merged with data.frames using `dplyr::*_join()`-functions. The `cut_to_classes()` is a wrapper for `cut()` -function and is used for categorizing data for maps with tidy labels.

```
mapdata <- get_eurostat_geospatial(nuts_level = 0) %>%
  right_join(dat_2015) %>%
  mutate(cat = cut_to_classes(values, n=4, decimals=1))
head(select(mapdata, geo, values, cat), 3)
## geo values cat geometry
## AT 1.49 1.5 ~< 1.6 MULTIPOLYGON (((15.54245 48...
## BE 1.70 1.6 ~< 1.8 MULTIPOLYGON (((5.10218 51...
## BG 1.53 1.5 ~< 1.6 MULTIPOLYGON (((22.99717 43...
```

Plot a Map

The sf-object returned are ready to be plotted with `ggplot::geom_sf()`-function.

```
ggplot(mapdata, aes(fill = cat)) +
  scale_fill_brewer(palette = "RdYlBu") +
  geom_sf(color = alpha("white", 1/3), alpha = .6) +
  xlim(c(-12, 44)) + ylim(c(35, 70)) +
  labs(title = "Total fertility rate, 2015",
       subtitle = "Avg. number of life births per woman",
       fill = "%")
```



This onepager presents the eurostat package 2014-2019
Leo Lahti, Janne Huovari, Markus Kainu, Przemyslaw Biecek
package version 3.3.55 URL: <https://github.com/rOpenGov/eurostat>

Retrieval and Analysis of Eurostat Open Data with the eurostat Package.
Leo Lahti, Janne Huovari, Markus Kainu, and Przemysław Biecek.
The R Journal, 9(1):385–392, 2017.





Access Eurostat data with eurostat::cheat sheet

Search and download

The `search_eurostat(pattern,...)` function scans the directory of Eurostat tables and returns codes and descriptions of tables that match pattern.

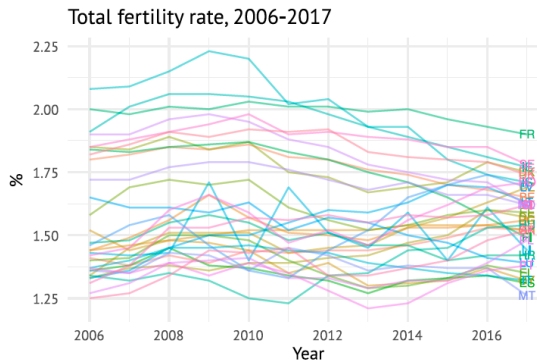
The `get_eurostat(id,time_format="date",filters="none",type="code",cache=TRUE,...)` function downloads the requested table from the Eurostat bulk download facility or from The Eurostat Web Services JSON API (if filters are defined).

Add labels

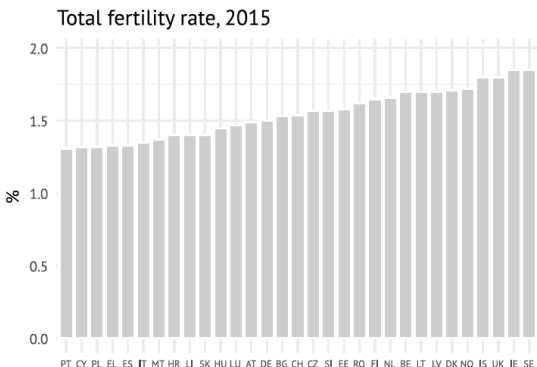
The `label_eurostat(x,lang="en",...)` gets definitions for Eurostat codes and replace them with labels in given language ('en','fr' or 'de')

eurostat and plots

The `get_eurostat()` function returns tibbles in the long format. Packages `dplyr` and `tidyr` are well suited to transform these objects. The `ggplot2` -package is well suited to plot these objects.



```
dat_2015 <- dat %>%
  filter(time == "2015-01-01")
ggplot(dat_2015, aes(x = reorder(geo, values), y = values)) +
  geom_col(color = "white", fill = "grey80") +
  theme(axis.text.x = element_text(size = 6)) +
  labs(title = "Total fertility rate, 2015",
       y = "%", x = NULL)
```



eurostat and maps

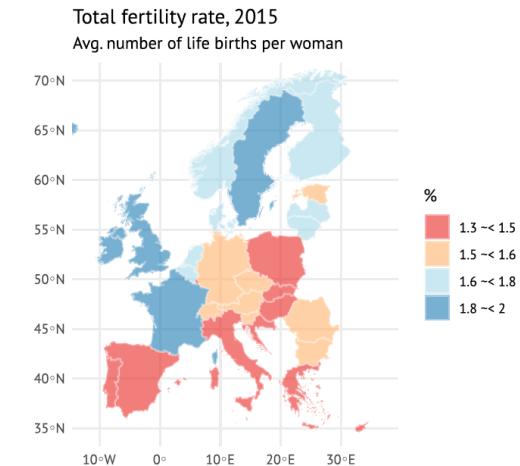
There are two functions to work with geospatial data from GISCO. The `get_eurostat_geospatial()` returns spatial data as `sf`-object. Object can be merged with `data.frames` using `dplyr::join()`-functions. The `cut_to_classes()` is a wrapper for `cut()` -function and is used for categorizing data for maps with tidy labels.

```
mapdata <- get_eurostat_geospatial(nuts_level = 0) %>%
  right_join(dat_2015) %>%
  mutate(cat = cut_to_classes(values, n=4, decimals=1))
head(select(mapdata, geo, values, cat), 3)
## geo values cat geometry
## AT 1.49 1.5 ~< 1.6 MULTIPOLYGON (((15.54245 48...
## BE 1.70 1.6 ~< 1.8 MULTIPOLYGON (((5.10218 51...
## BG 1.53 1.5 ~< 1.6 MULTIPOLYGON (((22.99717 43...
```

Plot a Map

The `sf`-object returned are ready to be plotted with `ggplot::geom_sf()`-function.

```
ggplot(mapdata, aes(fill = cat)) +
  scale_fill_brewer(palette = "RdYlBu") +
  geom_sf(color = alpha("white", 1/3), alpha = .6) +
  xlim(c(-12,44)) + ylim(c(35,70)) +
  labs(title = "Total fertility rate, 2015",
       subtitle = "Avg. number of life births per woman",
       fill = "%")
```



This onepager presents the eurostat package 2014-2019
Leo Lahti, Janne Huovari, Markus Kainu, Przemyslaw Biecek
package version 3.3.55 URL: <https://github.com/rOpenGov/eurostat>

Retrieval and Analysis of Eurostat Open Data with the eurostat Package.
Leo Lahti, Janne Huovari, Markus Kainu, and Przemyslaw Biecek.
The R Journal, 9(1):385–392, 2017.



Download the table

The **get_eurostat(id, time_format = "date", filters = "none", type = "code", cache = TRUE, ...)** function downloads the requested table from the Eurostat bulk download facility or from The Eurostat Web Services JSON API (if filters are defined). Downloaded data is cached (if cache=TRUE).

Additional arguments define how to read the time column (time_format) and if table dimensions shall be kept as codes or converted to labels (type).

```
ct <- c("AT", "BE", "BG", "CH", "CY", "CZ", "DE", "DK", "EE", "EL", "ES",  
       "FI", "FR", "HR", "HU", "IE", "IS", "IT", "LI", "LT", "LU", "LV",  
       "MT", "NL", "NO", "PL", "PT", "RO", "SE", "SI", "SK", "UK")  
dat <- get_eurostat(id="tps00199", time_format="num",  
                  filters = list(geo = ct))  
  
dat[1:2,]  
## indic_de geo      time values  
## TOTFERRT AT       2006      1.41  
## TOTFERRT AT       2007      1.38
```

.

```
tmp <- get_eurostat("cdh_e_fos")  
yy <- get_eurostat(id = "nrg_pc_202", filters = list(  
  geo = c("EU28", "FI"),  
  unit = "EUR_HAB",  
  indic_na = "B1GM"  
))
```

Eurostat

De unge

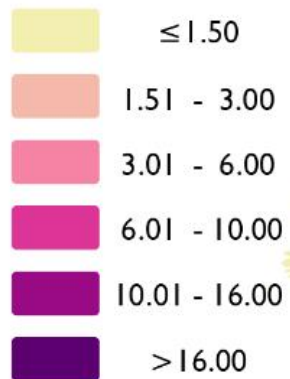
mødre



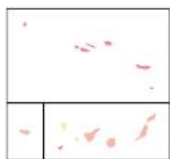
Teen moms

Landgeist.com
@Land_geist
@Landgeist

% of total live births
in 2020 where the
mother is aged <20



Note:
Data for the UK is from 2018.



Source: Eurostat, 2020

Highest

1. Sliven (30.18%)
2. Yambol (23.68%)
3. Vidin (18.60%)
4. Calarasi (18.50%)
5. Pazardzhik (17.90%)

Lowest

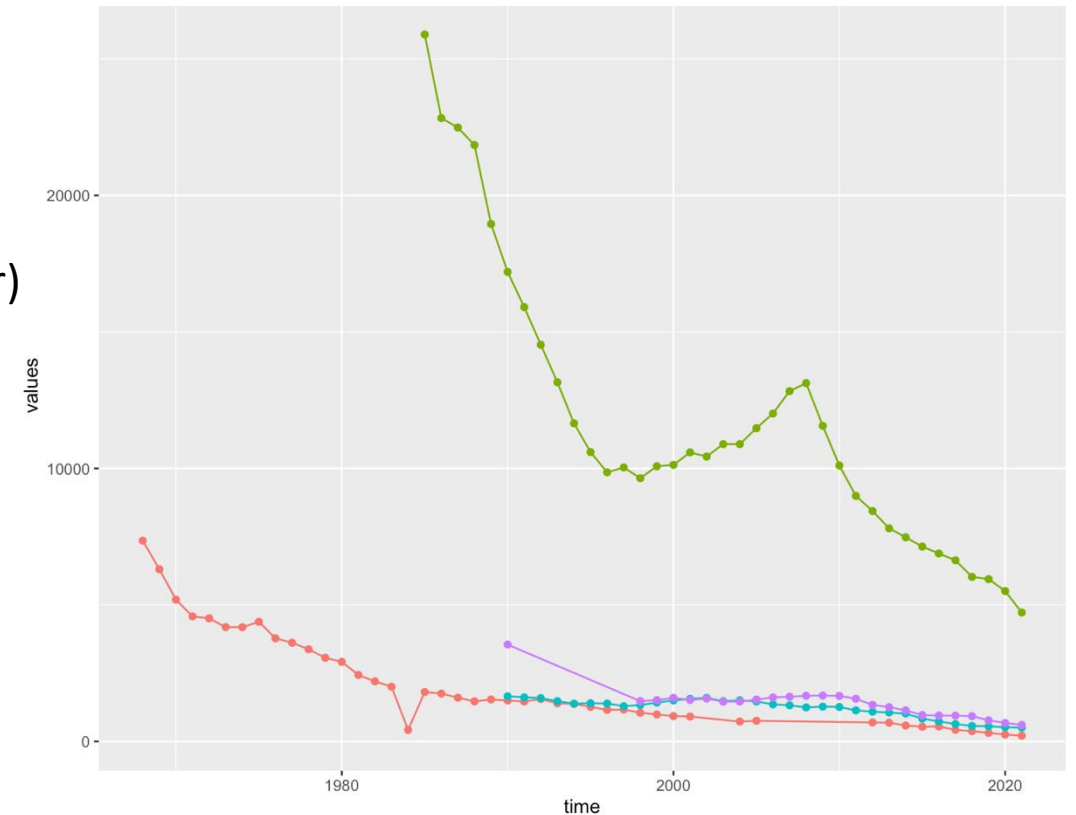
1. Evrytania (0.00%)
- Kastoria (0.00%)
- Lefkada (0.00%)
- Obwalden (0.00%)
- Orkney Islands (0.00%)
- Osttirol (0.00%)
- Schaffhausen (0.00%)



eurostat

Data-retrieval CASE: EU's unge mødre

1. Find tabel om fødsler
2. Eksperimenter med filtre
 1. Kig i koden og kørs eks
3. Hent alle data og inspicer obs-typer (variabler)
 1. Tjek for NA
 2. Filtrer så du kun har lande (ikke aggregerede observationer)
4. Tjek plot for teenagers i udvalgte lande
 1. Find de lande som du vil sammenligne med
 2. Find det range af årstal som dækker alle lande
5. Skal gøres relativt i forhold til start-værdien





Metadata I

Code lists - Metadata - Eurostat

ec.europa.eu/eurostat/web/metadata/code-lists

An official website of the European Union How do you know?

eurostat

Log in

Home Data News Publications About

Home > Data > Metadata > Code lists

METADATA

Code lists

- Overview
- Reference metadata reporting standards
- Classifications
- Code lists**
- Concepts and definitions
- National metadata

Code lists are

They are struc

consist of either all or a subset of items in a code list. Code lists are used for building and derived data products in the [Eurostat's database](#).

Eurostat currently uses around **600 code lists** to collect, store, and publish statistical

For example, the multidimensional dataset [Population on 1 January by age and sex](#) [demo_pjan] uses the code lists age, sex, geo, time, and unit.

Eurostat is continuously working towards the harmonisation of structural metadata and the

Access

All Eurostat code lists are available in three languages – English, French, and German – from [Eurostat's bulk download](#).

Eurostat standard code lists are also available in the [Euro SDMX registry](#), Eurostat's implementation of a statistical data and metadata exchange (SDMX) registry.

Statistics | Eurostat

ec.europa.eu/eurostat/databrowser/bulk?lang=en

eurostat Data Browser

Sign in | English EN

Search ...

ALL DATA RECENTLY UPDATED **DOWNLOADS** Info Help

Download operations
Perform operations on multiple items at the same time

Apply download operations on
Data **Code lists** Metadata Prodcom data Comext code lists
Comext metadata

Download operations
Select operations Apply

Code lists 1 item selected / 1 item

☒	Code	
☒	ORD_BRTH	

ESTAT_ORD_BRTH_1.0_EN.tsv

TOTAL	Total
0	No-one
1	First
2	Second
3	Third
4	Fourth
5	Fifth
GE4	Fourth and following
GE6	Sixth and following
UNK	Unknown



Metadata II

SDMX Registry

webgate.ec.europa.eu/sdmxregistry/

Euro SDMX Registry

About | Legal notice | Cookies | Contact | Help | Log in

European Commission > Eurostat > Euro SDMX Registry > Home

Repository Administration Data Registrations

☐ Include remote registries

search

What's new (607 updated items)

Agency Schemes [0]

Category Schemes [1]

Code lists [473]

Concept Schemes [7]

Constraints [23]

Dataflows [64]

Data Consumer Schemes [0]

Data Provider Schemes [0]

Data Structures [23]

Hierarchical Code lists [0]

Metadataflows [13]

Metadata Structures [3]

Organisation Unit Schemes [0]

Processes [0]

Provision Agreements [0]

Structure Sets [0]

Welcome to Euro SDMX Registry

Euro SDMX Registry is Eurostat's implementation of the SDMX Registry specifications as published by the SDMX initiative sdmx.org. This implementation is based on SDMX 2.1 (as published on April 2011) but is also capable of importing and exporting SDMX 2.0 artefacts. Euro SDMX Registry allows browsing, searching, editing and subscribing to artefacts according to the SDMX rules. Moreover, an advanced access control mechanism, allows for distributed maintenance of artefacts controlling also their visibility.

Currently, the following SDMX artefacts are supported:

Agency Schemes

Category Schemes

Code lists

Classifications

Concept Schemes

Constraints

Dataflows

Data Consumer Schemes

Data Provider Schemes

Data Structures (aka DSDs, or Key Families)

Hierarchical Code lists

Metadataflows

Metadata Structures (aka MSDs)

Organisation Unit Schemes

Provision Agreements

Processes

Structure Sets



European Union (EU)

Belgium	(BE)	Greece	(EL)	Lithuania	(LT)	Portugal	(PT)
Bulgaria	(BG)	Spain	(ES)	Luxembourg	(LU)	Romania	(RO)
Czechia	(CZ)	France	(FR)	Hungary	(HU)	Slovenia	(SI)
Denmark	(DK)	Croatia	(HR)	Malta	(MT)	Slovakia	(SK)
Germany	(DE)	Italy	(IT)	Netherlands	(NL)	Finland	(FI)
Estonia	(EE)	Cyprus	(CY)	Austria	(AT)	Sweden	(SE)
Ireland	(IE)	Latvia	(LV)	Poland	(PL)		

European Free Trade Association (EFTA)



eurostat

Data-retrieval CASE: EU's unge mødre

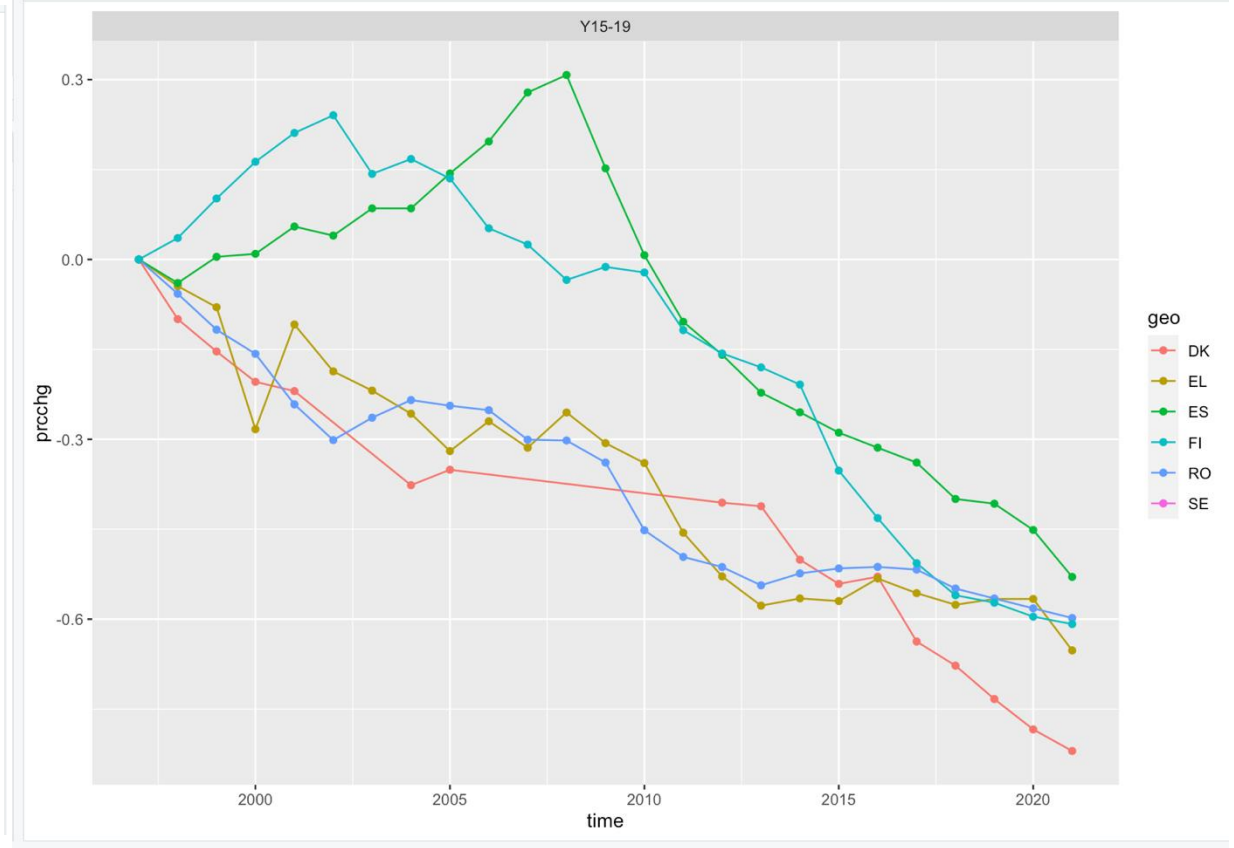
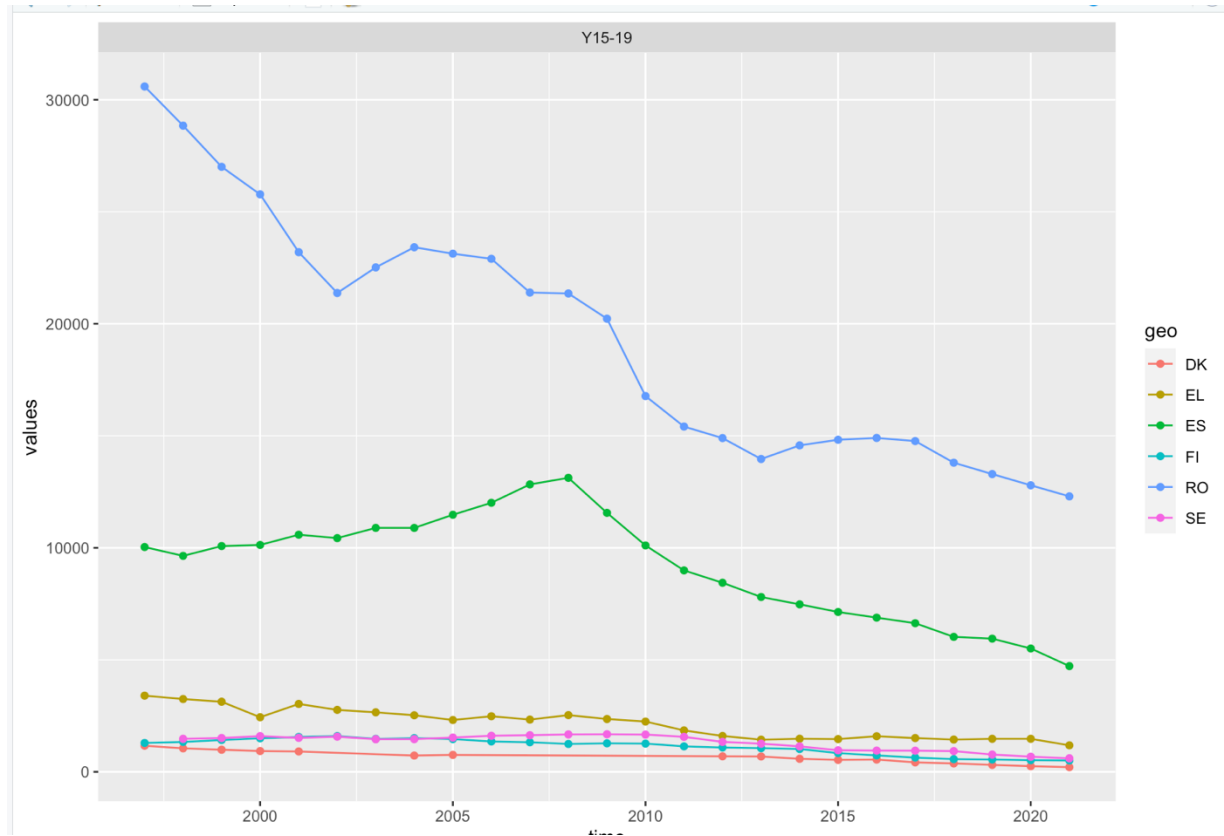
1. Find tabel om fødsler
2. Eksperimenter med filtre
 1. Kig i koden

```
#' dd <- get_eurostat("nama_10_gdp",  
#' filters = list(  
#'   geo = "FI",  
#'   na_item = "B1GQ",  
#'   unit = "CLV_I10"  
#' )  
#'
```

		freq	unit	na_item	geo	time	values
1	A		CLV_I10	B1GQ	FI	1975-01-01	NA
2	A		CLV_I10	B1GQ	FI	1976-01-01	NA
3	A		CLV_I10	B1GQ	FI	1977-01-01	NA
4	A		CLV_I10	B1GQ	FI	1978-01-01	NA
5	A		CLV_I10	B1GQ	FI	1979-01-01	NA
6	A		CLV_I10	B1GQ	FI	1980-01-01	48.935
7	A		CLV_I10	B1GQ	FI	1981-01-01	49.579
8	A		CLV_I10	B1GQ	FI	1982-01-01	51.120
9	A		CLV_I10	B1GQ	FI	1983-01-01	52.713
10	A		CLV_I10	B1GQ	FI	1984-01-01	54.419
11	A		CLV_I10	B1GQ	FI	1985-01-01	56.351
12	A		CLV_I10	B1GQ	FI	1986-01-01	57.903



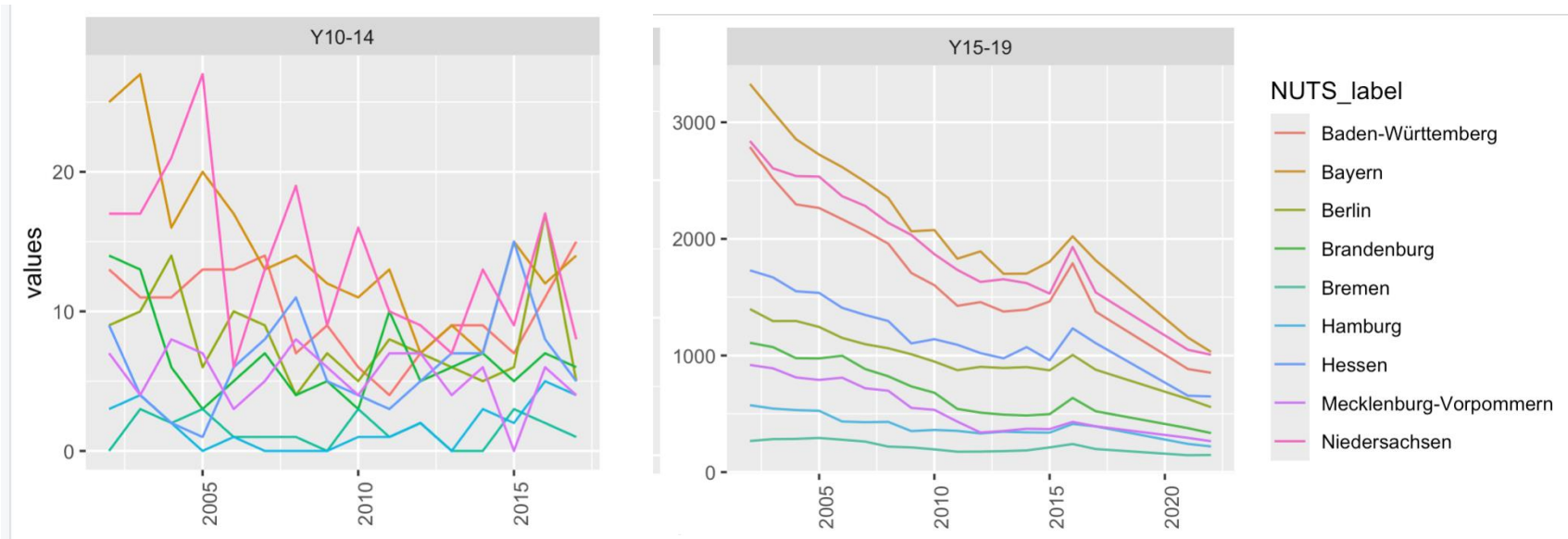
demo_fordagec





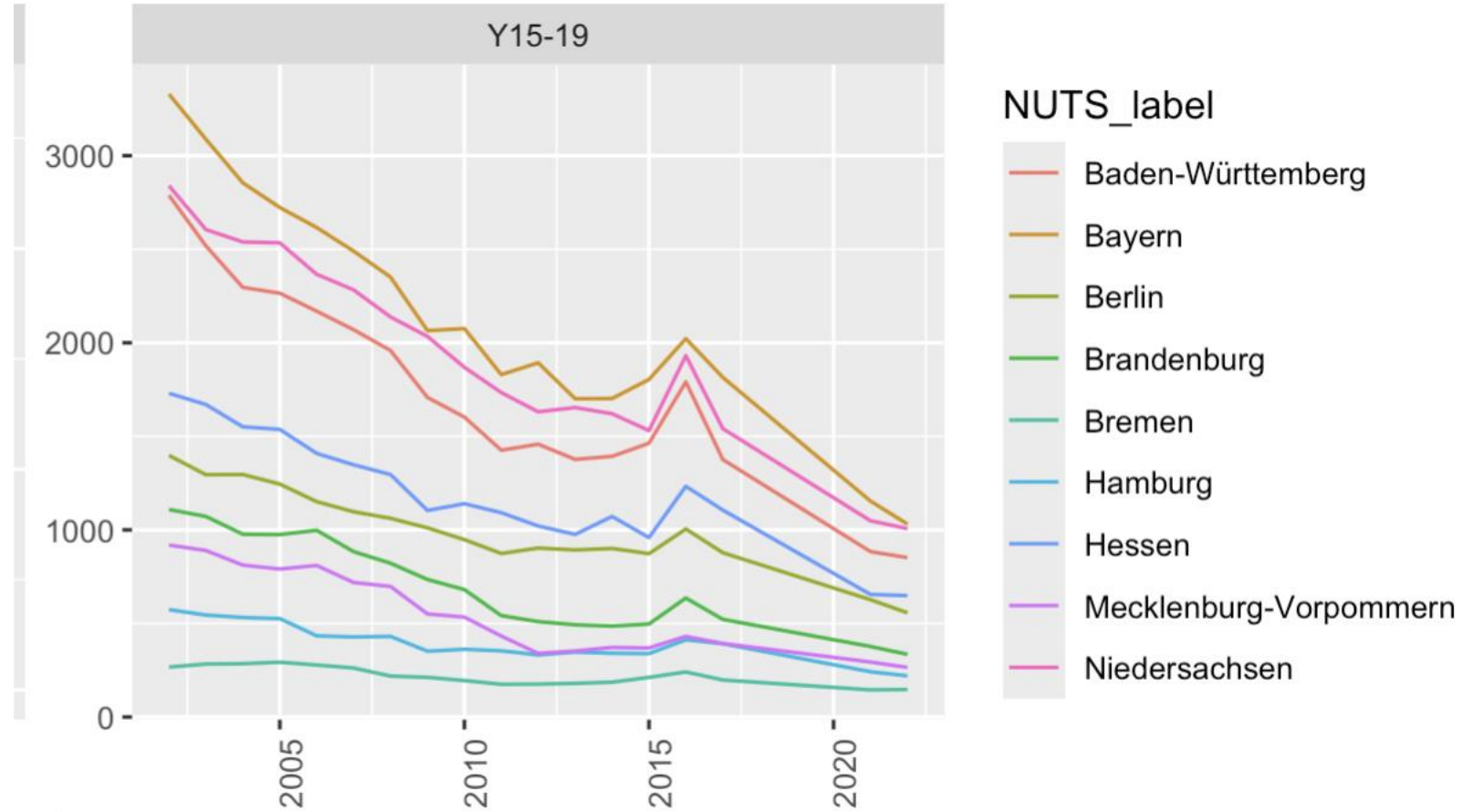
Teen moms in Germany

1. Der er noget helt galt med mit plot (ud over at det plot-teknisk er dårligt). Hvad er det?
2. Hvilken tabel skal I hente for at kunne genskabe plottet "Teen moms" fra mit slide ovenfor?
3. Find en tabel i Eurostat som kan vise fordelingen af teen-moms i de tyske hovedregioner. Sørg for at tjekke metadata inden (`get_eurostat_dsd`) så I ved hvilke variabler og hvilke værdier I har adgang til.
4. Find en anden kilde i Eurostat hvor I kan trække navnene på NUTS-koderne så de kan bruges i plottet. Det vil sige at I skal merge to data-kilder.



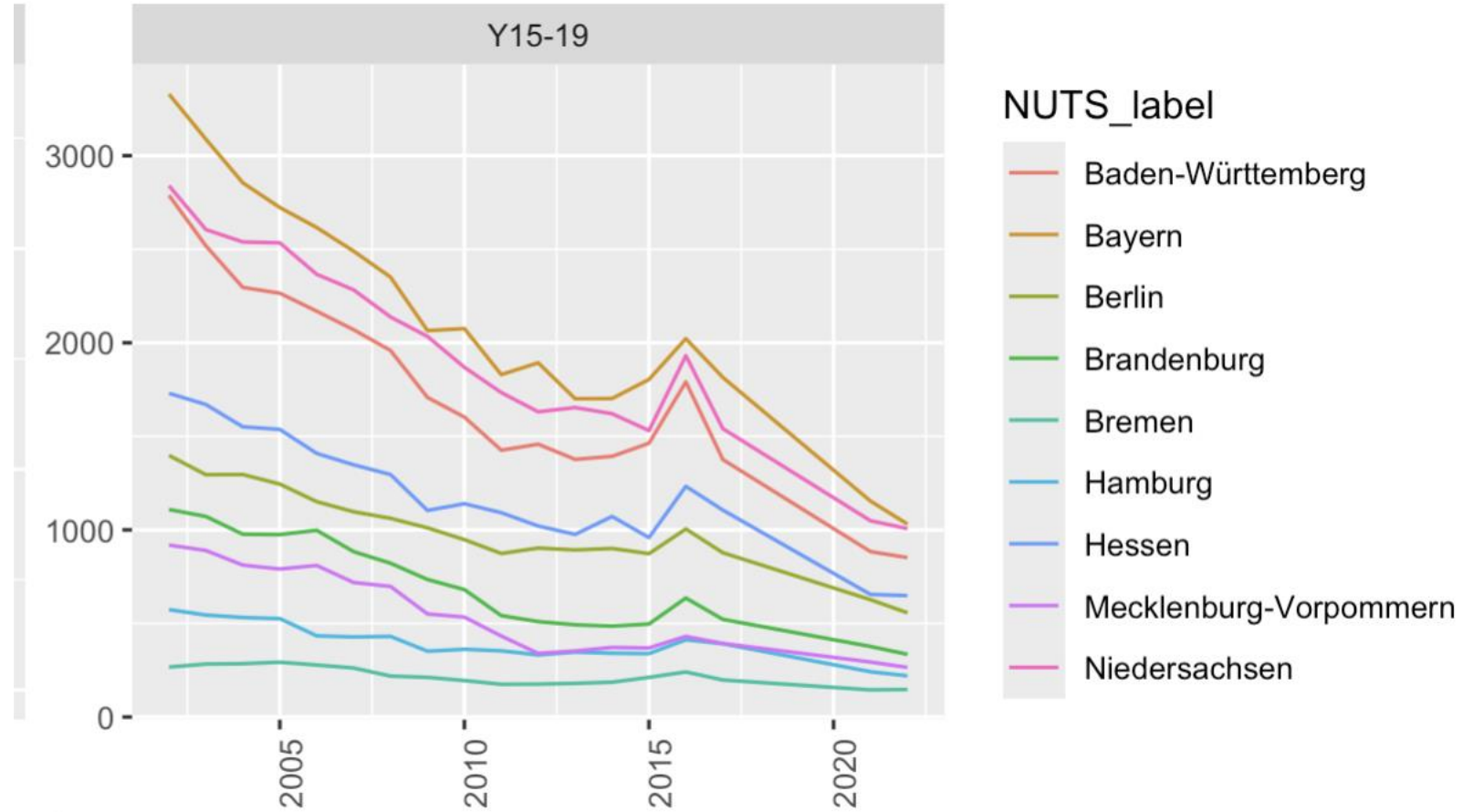


Find et data-sæt som kan beskrive fordelingen af unge mødre i tyske regioner



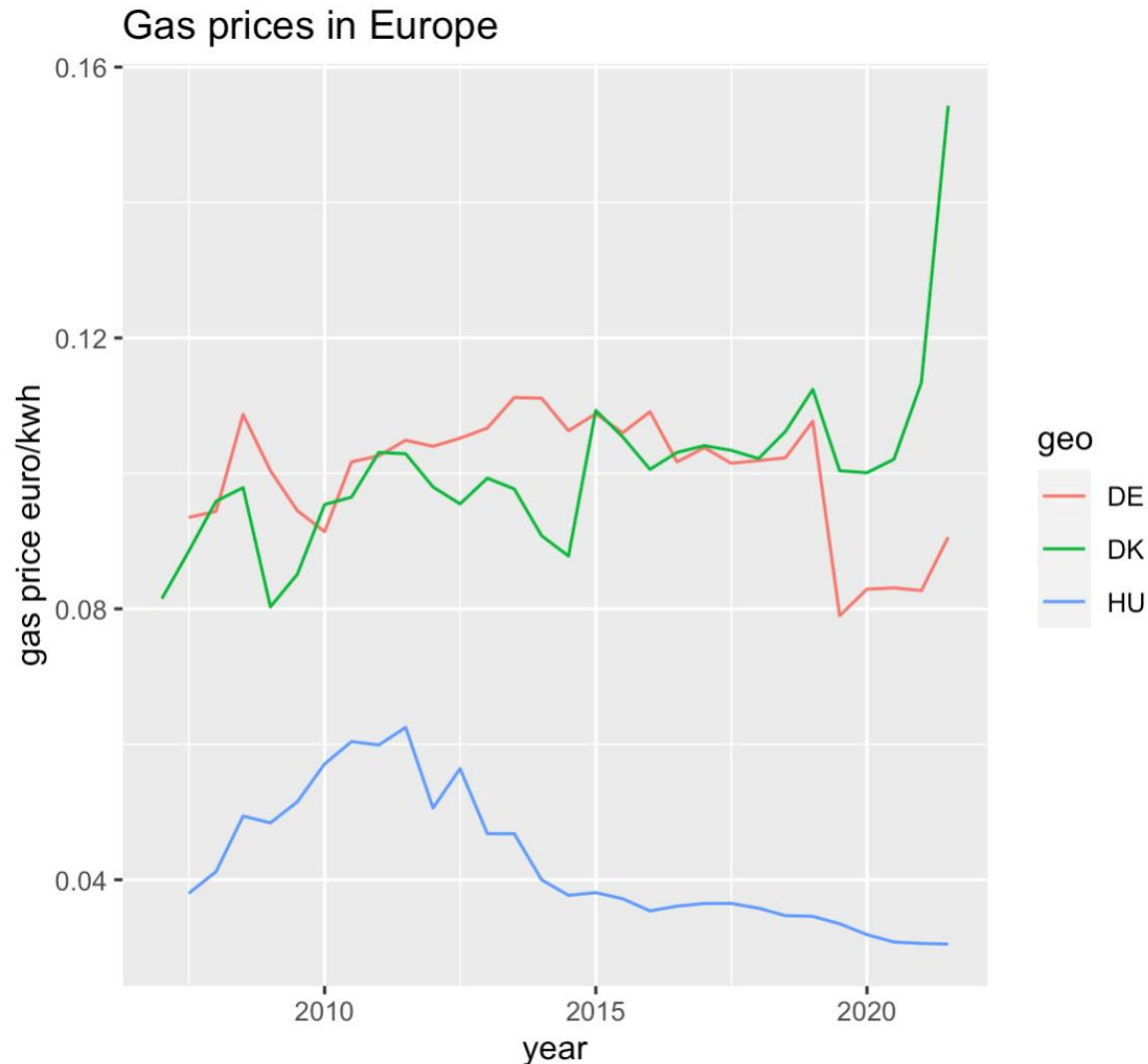


Find et data-sæt som kan beskrive fordelingen af unge mødre i tyske regioner





Genskab denne graf ud fra **nrg_pc_202**



- 1) Find data og meta-data via UI
- 2) Hent meta-data via **get_eurostat_dsd**
- 3) Undersøg filtreringsmulighederne på <https://github.com/eurostat/restatapi>
 - 1) Hent data fra før 2017, enhed KWH og lande er Danmark og Ungarn
- 4) Hent alle data og filtrer i R så du får noget a la grafen til venstre